Train a Human Neural Network KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

The rule. total = ($A \times$ Attack weight) + ($S \times$ Special weight) + bias. More than 0: **FIGHTING**. 0 or less: **PSYCHIC**.

The fix rule. Right: no change. Said P, really F: **ADD** A, S, 1. Said F, really P: **SUBTRACT** A, S, 1.

Pass 1. Starting weights: Attack −1, Special +1, bias 0.

Card	A	S	Total	Says	Really	Right?	Attack w	Special w	Bias
T1 Machop	3	1	−2	P	F	no	2	2	1
T2 Abra	1	4	11	F	P	no	1	−2	0
T3 Machamp	4	2	0	P	F	no	5	0	1
T4 Espeon	2	4	11	F	P	no	3	−4	0
T5 Hitmonlee	4	1	8	F	F	yes	3	−4	0
T6 Kirlia	1	2	−5	P	P	yes	3	−4	0
T7 Conkeldurr	5	2	7	F	F	yes	3	−4	0
T8 Alakazam	2	5	−14	P	P	yes	3	−4	0
T9 Makuhita	2	1	2	F	F	yes	3	−4	0
T10 Mr. Mime	2	3	−6	P	P	yes	3	−4	0

Mistakes in pass 1: 4

Pass 2. Same cards, same order. Start with your weights from the end of pass 1.

Card	Total	Says	Right?
T1 Machop	5	F	yes
T2 Abra	−13	P	yes
T3 Machamp	4	F	yes
T4 Espeon	−10	P	yes
T5 Hitmonlee	8	F	yes

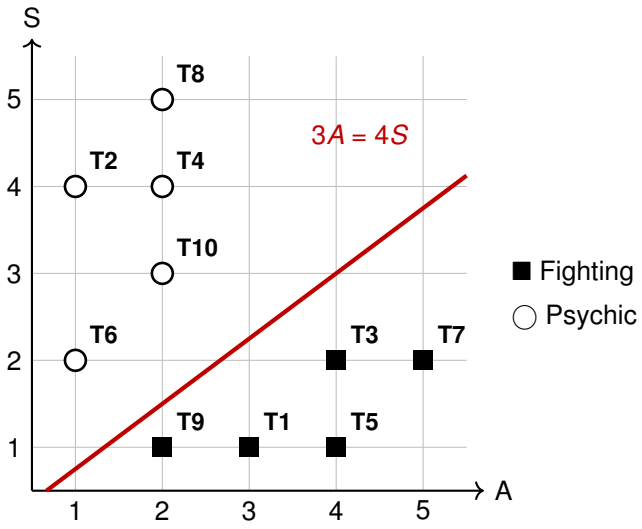
Card	Total	Says	Right?
T6 Kirlia	−5	P	yes
T7 Conkeldurr	7	F	yes
T8 Alakazam	−14	P	yes
T9 Makuhita	2	F	yes
T10 Mr. Mime	−6	P	yes

Mistakes in pass 2: 0 (If you made mistakes, do a pass 3 on the back.)

Test cards. Do **not** fix the weights on a test! Final weights: Attack 3 Special −4 bias 0

Card	A	S	Total	Says	Really (flip the card)
X1 Primeape	4	2	4	F	Fighting
X2 Mewtwo	4	5	−8	P	Psychic
X3 Hypno	2	2	−2	P	Psychic
X4 Gallade	4	2	4	F	Psychic and Fighting
X5 Lucario	4	4	−4	P	Fighting and Steel
X6 Medicham	2	2	−2	P	Fighting and Psychic

Talk about it. The neuron said Gallade is Fighting. Gallade is Psychic **and** Fighting. Was the neuron wrong?

The neuron **was** / **was not** wrong about Gallade, because Answers vary.

Go Further (Day 2). The neuron's line. The training cards are on the graph. Draw the line where the total is 0.